Zhichen Zeng

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Research Interest

LLMs/VLMs & Agents; Geometric ML; Recommender system; Trustworthy ML

Education

University of Illinois Urbana-Champaign — IL, USA
Ph.D. in Computer Science; GPA: 4.0/4.0 — *Aug. 2023 – Dec. 2026*

University of Illinois Urbana-Champaign — IL, USA
M.S. in Computer Science; GPA: 4.0/4.0 — *Aug. 2021 – May. 2023*

Zhejiang University — Zhejiang, China
B.Eng. in Automation; GPA: 3.9/4.0 — *Sept. 2016 – Jun. 2020*

Internship

Amazon — CA, USA
Applied Scientist Intern with Huiyuan Chen, Chaosheng Dong, and Haiyang Zhang — *May 2026 – Aug. 2026*
Unlocking multi-agent synergy in a single agent system. Paper currently under review.

Meta — CA, USA
Research Scientist Intern with Mengyue Hang, and Xiaoyi Liu — *May 2025 – Aug. 2025*
Efficient model scaling via hierarchical LoRA MoE. Paper currently under reivew.

Meta — CA, USA
Research Scientist Intern with Jiyan Yang — *May 2024 – Nov. 2024*
Heterogeneous Recsys model scaling (InterFormer). Model deployed as Meta's Ads foundation model.

ByteDance — Beijing, China
Machine Learning Engineer Intern with Qile Xu and Xiuying Zhao — *Jul. 2020 – Jun. 2021*
Ads bidding strategy optimization.

Publications

LLM/VLM/Agent

1. **Zeng Z**, Bao W, Lin X, Qiu R, Wei T, Ning X, Yan Y, Luo C, Cheng X, He J, Tong H. Subspace Alignment for Vision-Language Model Test-time Adaptation. EMNLP 2026 Main (AR: 15.4%).
2. **Zeng Z***, Yu Q*, Lin X, Qiu R, Ning X, Wei T, Yan Y, He J, Tong H. Harnessing Model Consistency for Robust Test-Time LLM Ensemble. EACL 2026 Findings (AR: 16.1%).
3. **Zeng, Z**, Chen H, Cheng J, Zha J, Liu M, Chen Y, Yang X, Dong C, Zhang H, and Tong H. One Model, Many Minds: Unlocking Multi-Agent Synergy in a Single Agent via Mixture of Roles. Preprint 2026.
4. Wei T, **Zeng Z**, Qiu R, Liu Z, Ning X, He X, Bao W, Lin X, He Q, Tang X, Tong H, He J. Inference Scaling of LLM Ensembling: Bridging Token Spaces with Token Translation. COLM 2026 (AR: 29%).
5. UIUC, Meta, Amazon, Google Deepmind, UCSD, Yale. Agentic Reasoning for Large Language Models. TMLR 2026.
6. UIUC, Meta, Stanford. Code as Agent Harness. Preprint 2026.
7. Lin X, Li P, **Zeng Z**, Li T, Wei T, Ning X, Li G, Chen Y, Tong H. ALERT: Zero-shot LLM Jailbreak Detection via Internal Discrepancy Amplification. EMNLP 2026 Main (AR: 15.4%).
8. Cui C, Wei T, Chen Z, Qiu R, **Zeng Z**, Liu Z, Ning X, Zhou D, He J. AdaFuse: Adaptive Ensemble Decoding with Test-Time Scaling for LLMs. ACL 2026 Main (AR: 19%).
9. Qiu R, Zeng H, Xia Y, Meng Y, Chen R, Feng J, Fu D, Wang Q, Liu J, Xiao J, Fan X, Zhang B, Li H, Liu Z, Yoo H, **Zeng Z**, Wei T, Tong H. Reinforcement Routing for Mixtures of LoRAs in Parameter-Efficient LLM Finetuning. COLM 2026 (AR: 29%).
10. Zhang Y, Yu D, Ge T, Song L, **Zeng Z**, Mi H, Jiang N, Yu D. Improving LLM General Preference Alignment via Optimistic Online Mirror Descent. NeurIPS 2025 Spotlight (AR: 24.52%).
11. Ai M, Wei T, Chen Y, **Zeng Z**, Zhao R, Varatkar G, Rouhani G, Tang X, Tong H, He J. ResMoE: Parameter Compression for Pre-Trained Mixture-of-Experts Transformer via Residual Restoration. KDD 2025 (AR: 19%).
12. Jing B, Chen S, Zheng L, Liu B, Li Z, Zou J, Wei T, Liu Z, **Zeng Z**, Qiu R, Lin X, et al. TSAQA: Time Series Analysis Question And Answering Benchmark. ACL Workshop on Generation, Evaluation and Metrics.

13. Yu Q, **Zeng Z**, Tieu K, Yang X, Qiu R, Yan Y, Liu L, Zhao Y, Chen L, He J, Tong H. From Inference to Adaptation: A Unified Optimal Transport View of Vision Language Model. Preprint 2026.
14. Yu Q, Qiu R, **Zeng Z**, Ning X, Zhao Y, Fu D, Xia Y, Li H, Tong H. Hierarchical Credit Assignment for RLVR on Fused Gromov-Wasserstein Geometry. Preprint 2026.
15. Jin K, Yan Y, **Zeng Z**, Lin X, Liu L, Xia Y, Tong H. Towards a Learnable Retriever for Textual Graph Question Answering. Preprint 2026.
16. Chen L, Qiu R, Yuan S, Liu Z, Wei T, Yoo H, **Zeng Z**, Yang D, Tong H. WAPITI: A Watermark for Finetuned Open-Source LLMs. Preprint 2024.

Geometric ML & Trustworthy ML

17. **Zeng Z**, Qiu R, Bao W, Wei T, Lin X, Yan Y, Abdelzaher TF, Han J, Tong H. Pave Your Own Path: Graph Gradual Domain Adaptation on Fused Gromov-Wasserstein Geodesics. TMLR 2026.
18. **Zeng Z**, Qiu R, Xu Z, Liu Z, Yan Y, Wei T, Ying L, He J, Tong H. Graph Mixup on Approximate Gromov-Wasserstein Geodesics. ICML 2024 (AR: 27.5%).
19. **Zeng Z**, Du B, Zhang S, Xia Y, Liu Z, Tong H. Hierarchical Multi-Marginal Optimal Transport for Network Alignment. AAAI 2024 (AR: 23.8%).
20. **Zeng Z**, Zhu R, Xia Y, Zeng H, Tong H. Generative Graph Dictionary Learning. ICML 2023 (AR: 27.9%).
21. **Zeng Z**, Zhang S, Xia Y, Tong H. PARROT: Position-Aware Regularized Optimal Transport for Network Alignment. theWebConf 2023 (AR: 19.2%).
22. Yu Q*, **Zeng Z***, Yan Y, Ying L, Srikantm R, Tong H. Joint Optimal Transport and Embedding for Network Alignment. theWebConf 2025 (AR: 19.8%).
23. Yu Q, **Zeng Z**, Yan Y, Liu Z, Jing B, Qiu R, Azad A, Tong H. PLANETALIGN: A Comprehensive Python Library for Benchmarking Network Alignment ICLR 2026 (AR: 28.0%).
24. Bao W, **Zeng Z**, Liu Z, Tong H, He J. Matcha: Mitigating Graph Structure Shifts with Test-Time Adaptation. ICLR 2025. (AR: 32.08%).
25. Yu Q, Qiu R, **Zeng Z**, Thai M, Liu H, Tong H. AvAtar: Learning to Align via Active Optimal Transport. ICML 2026 (AR: 26.6%).
26. Zheng L, Jing B, Li Z, **Zeng Z**, Wei T, Ai M, He X, Liu L, Fu D, You J, Tong H. PyG-SSL: A Graph Self-Supervised Learning Toolkit. CIKM 2025 (AR: 43.2%).
27. Liu Z, Qiu R, **Zeng Z**, Yoo H, Zhou D, Xu Z, Zhu Y, Weldemariam K, He J, Tong H. Class-Imbalanced Graph Learning without Class Rebalancing. ICML 2024 (AR: 27.5%).
28. Xu H, Yan Y, Wang D, Xu Z, **Zeng Z**, Abdelzaher TF, Han J, Tong H. SLOG: An Inductive Spectral Graph Neural Network Beyond Polynomial Filter. ICML 2024 (AR: 27.5%).
29. Yan Y, Hu Y, Zhou Q, Liu L, **Zeng Z**, Chen Y, Pan M, Chen H, Das M, Tong H. PACER: Network Embedding From Positional to Structural. theWebConf 2024 (AR: 21.1%).
30. Xu Z, Qiu R, Chen Y, Chen H, Fan X, Pan M, **Zeng Z**, Das M, Tong H. Discrete-state Continuous-time Diffusion for Graph Generation. NeurIPS 2024 (AR: 25.8%).
31. Liu Z, Qiu R, **Zeng Z**, Zhu Y, Hamann H, Tong H. AIM: Attributing, Interpreting, Mitigating Data-encoded Unfairness. KDD 2024 (AR: 20.0%).
32. Lin X, **Zeng Z**, Wei T, Liu Z, Tong H. CATS: Mitigating Correlation Shift for Multivariate Time Series Classification. Preprint 2025.
33. Yan Y, Chen Y, Chen H, Li X, Xu Z, **Zeng Z**, Liu Z, Tong H. THeGCN: Temporal Heterophilic Graph Convolutional Network. Preprint 2024.

Recommender System

34. **Zeng Z***, Liu X*, Hang M*, Liu X*, Zhou Q, Yang C, Liu Y, Ruan Y, Chen L, Chen Y, Hao Y, Xu J, Nie J, Liu X, Zhang B, Wen W, Yuan S, Wang K, Chen WY, Han Y, Li H, Yang C, Long B, Yu PS, Tong H, Yang J. InterFormer: Effective Heterogeneous Interaction Learning in CTR Prediction. CIKM 2025 (AR: 30%).
35. **Zeng Z**, Hang M, Liu X, Liu X, Lin X, Qiu R, Wei T, Liu Z, Yuan S, Yang C, Liu Y, Yin H, Yang J, Tong H. Hierarchical LoRA MoE for Efficient CTR Model Scaling. Preprint 2025.
36. Lin X, Tang Z, Cong W, Hang M, Wang K, Wang Y, **Zeng Z**, Li T, Yoo H, Liu Z, Ning X, Qiu R, Chen W, Chang S, Jin R, Li H, Tong H. Mixture of Sequence: Theme-Aware Mixture-of-Experts for Long-Sequence Recommendation. theWebConf 2026 (AR: 20.1%).
37. Liang M, Liu X, Li H, Jin R, Liu B, Suo Q, Zhou Q, Zhou S, Chen L, Zheng H, Li Z, Jiang S, Yang J, Xia X, Yang F, Badr Y, Wen E, Xu S, Chen H, Zhang Z, Nie J, Yang C, **Zeng Z**, Zhang W, Huang X, Li Q, Wang S, Lyu E, Lu W, Zhang R, Wang W, Rudy J, Hang M, Wang K. External Large Foundation Model: How to Efficiently Serve Trillions of Parameters for Online Ads Recommendation. theWebConf 2025 (AR: 19.8%).

38. Yoo H, **Zeng Z**, Kang J, Qiu R, Zhou D, Liu Z, Wang F, Xu C, Chan E, Tong H. Ensuring User-side Fairness in Dynamic Recommender Systems. theWebConf 2024 (AR: 21.1%).
39. Liu X, **Zeng Z**, Liu X, Yuan S, Song W, Hang M, Liu Y, Yang C, Kim D, Chen WY, Yang J, Han Y, Jin R, Long B, Tong H, Yu PS. A Collaborative Ensemble Framework for CTR Prediction. Preprint 2024.

Others

40. Li Z*, **Zeng Z***, Lin X*, Fang F, Qu Y, Xu Z, Liu Z, Ning X, Wei T, Liu G, Tong H. Flow Matching Meets Biology and Life Science: A Survey. Nature Portfolio Journal (npj) Artificial Intelligence.
41. **Zeng Z**, Ni D, Xiao G. Real-time heliostat field aiming strategy optimization based on reinforcement learning. Applied Energy (impact factor: 11.0).
42. **Zeng Z**, Ni D. Harvesting optimal operation strategies from historical data for solar thermal power plants using reinforcement learning. SolarPACES 2020 (Oral).

Honors and Awards

Dissertation Completion Fellowship, University of Illinois Urbana-Champaign	2026
Outstanding Teaching Assistant Award, University of Illinois Urbana-Champaign	2026
Exceptional Technical Accomplishment, Assistant Secretary of Defense for Critical Technologies	2024
Top Reviewer of NeurIPS 2023	2023
Conference Presentation Award, University of Illinois Urbana-Champaign	2023
Outstanding Graduates, Zhejiang University	2020
Provincial Scholarship, Zhejiang Province	2018, 2019
Merit Student, Zhejiang University	2018, 2019
Second-class Scholarship, Zhejiang University	2018, 2019
Meritorious Winner of Mathematical Contest In Modelling, COMAP, USA	2019
First-class Research and Innovation Scholarship, Zhejiang University	2018
Three-star Volunteer, Zhejiang University	2018
Outstanding Student Leader, Zhejiang University	2018
First Prize of Physical Innovation Competition, Zhejiang Province	2018
Second Prize of Academic Physics Competition, Zhejiang University	2018
Second Prize of Mathematical Modeling Competition, Zhejiang University	2018
Third Prize of Higher Mathematics Competition, Zhejiang Province	2017
First Prize of Chinese Physics Olympiad, Chinese Physics Society	2015

Professional Services

Conference reviewer: ICML, NeurIPS, ICLR, WWW, KDD, AAAI, IJCAI, CIKM, IEEE Big data
Journal reviewer: TPAMI, TKDD, TBD, TIST, TCBB

Projects/Fundings

Modeling Adversarial Activities	*DARPA (HR001121C0165)*
A GPU/CPU Hybrid Cluster for Accelerating Network-of-X Research and Education	*ARO (W911NF2110088)*
Understanding Social Network-Transcendent Online/Offline Behavioral Dynamics	*AFOSR (FA9550-24-1-0002)*
Center of Excellence for Accelerating Operational Efficiency (CAOE)	*DHS (17STQAC00001-07-00)*
Towards A Computational Foundation of Teams Network Science	*NSF2433308*
NetSafe: Towards a Computational Foundation of Safe Graph Neural Networks	*NSF2416070*
Reconstruction of Diffusion History in Cyber and Human Networks with Applications in Epidemiology and Cybersecurity	*NSF 2324770*
General-Purpose Scalable Technologies for Fundamental Graph Problems	*NSF2316233*
Towards a Theoretic Foundation for Optimal Deep Graph Learning	*NSF2134079*
FAI: Towards a Computational Foundation for Fair Network Learning	*NSF1939725*
Machine Learning for Soil Health	*NIFA (2020-67021-32799)*
Multi-dimensional User Understanding on Graphs	*Meta*